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# Adaptive backstepping trajectory tracking control without velocity measurements for quadrotor UAVs under multiple unknown disturbances

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**Abstract**
An unmanned aerial vehicle (UAV) quadrotor, subject to various unknown disturbances, must be capable of performing both aggressive and non-aggressive maneuvers. Therefore, this paper presents an adaptive backstepping trajectory tracking control technique without velocity measurements for quadrotor UAVs subject to multiple nonlinear disturbances. Despite the quadrotor's underactuated dynamical model, this paper explores a fully actuated controller for trajectory tracking in a 3D plane. Initially, a tracking error is defined, and the dynamic surface method is employed to derive a filter and auxiliary controller, addressing the requirement for repeated derivatives of the control input. Subsequently, a Lyapunov candidate function is introduced to derive adaptive laws for tuning the control parameters of the auxiliary controller and filter, ensuring stability of the error dynamics. Additionally, a high-gain observer is designed for state estimation and integrated with the controller to achieve the control objectives. For disturbance attenuation, a disturbance observer is designed and integrated into the control scheme for estimation and rejection of von Kármán turbulence and noises modeled by Gaussian distribution and uniformly random distribution. Stability analysis, based on Lyapunov criteria, and simulation study, using the DJI F450 quadrotor, validate the effectiveness and stability of the proposed controller.

**Keywords** Unmanned aerial vehicle · Robust control · Aggressive maneuvers · Turbulence · Disturbance observer

## 1 Introduction

Quadrotor control presents a significant challenge in the research community, primarily due to its underactuated, multi-input, multi-output (MIMO) dynamic system. Achieving control requires the design of four control inputs to regulate six outputs, encompassing both rotational and translational dynamics. The basic mechanical structure of a quadrotor consists of four rotors, which are mounted at the ends of the airframe's arms. These configurations can be either "X" or "+" shaped, as depicted in Fig. 1 [1]. Each rotor generates thrust, and by manipulating the thrust levels of each rotor, a range of flight modes can be attained, including vertical takeoff and landing (VTOL) as well as trajectory tracking. The rotor speeds and ON-OFF control topology required to execute specific flight missions are summarized in Table 1. Due to their versatility and high maneuverability, quadrotor unmanned aerial vehicles (UAVs) have found widespread use across military, commercial, industrial, and civilian applications. Beginning with basic VTOL control objectives and extending to aggressive and non-aggressive maneuvers, researchers have conducted extensive studies to improve and develop novel control techniques.

Traditional and advanced linear control techniques have been widely applied to quadrotor control systems to achieve stabilization, hovering, and trajectory tracking in the Cartesian plane [2]. A proportional–integral–derivative (PID) control strategy has been developed for rotational control, with the objective of enhancing energy efficiency by optimizing the PID gain parameters using a differential evolution algorithm [3]. To address both trajectory tracking and power efficiency, [4] proposed a PID controller integrated with the cuckoo search algorithm, which significantly improved the controller's ability to manage parametric uncertainties. Sim-

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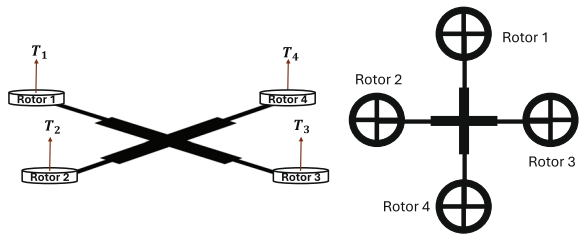


**Fig. 1** "×" and "+" configuration of quadrotor

**Table 1** Rotor Speeds for "×"—Quadrotor Movements

| Movement | $T_1$ | $T_2$ | $T_3$ | $T_4$ |
|---|---|---|---|---|
| **Vertical Lift (VTOL)** | ON | ON | ON | ON |
| **Pitch Forward** | OFF | ON | ON | OFF |
| **Pitch Backward** | ON | OFF | OFF | ON |
| **Roll Left** | OFF | ON | ON | OFF |
| **Roll Right** | ON | OFF | OFF | ON |
| **Yaw Clockwise** | OFF | ON | OFF | ON |
| **Yaw Counterclockwise** | ON | OFF | ON | OFF |

ilarly, [5] introduced an implicit PID controller designed to achieve robust tracking control under the influence of external disturbances, utilizing linear matrix inequalities (LMIs) to establish system stability and convergence properties. However, these approaches either focus exclusively on the 3-degree-of-freedom (DOF) rotational control problem or fail to account for disturbances during full trajectory tracking. Moreover, during the modeling process, important factors such as Coriolis forces, hub moments, and other forces inherent in the Euler–Lagrange formulation are often neglected.

The linear quadratic regulator (LQR) control technique has been employed to achieve 3-degree-of-freedom (DOF) control for tracking the Euler angles of quadrotors [6]. This method incorporates integral action within a differential game framework to eliminate steady-state errors and enhance control performance. While LQR offers a straightforward implementation and robust performance, it presents challenges in selecting appropriate weighting matrices. To address this issue, optimization techniques such as genetic algorithms and differential evolution have been applied to improve trajectory tracking performance [7]. Furthermore, the linear quadratic Gaussian (LQG) controller has been proposed to optimize control gains and select the appropriate weighting matrices, thereby ensuring precise object tracking and safe landing of quadrotor UAVs [8]. Despite achieving commendable control performance, these methods are highly dependent on accurate mathematical models of the quadrotor system and are sensitive to uncertainties and parametric variations [9]. As a result, adaptive control strategies for quadrotor systems have garnered considerable attention in recent research.

Adaptive control is an online estimation technique used to assist in tuning control gains and has been widely applied in quadrotor UAV control applications. Adaptive criteria for adjusting the weighting matrices in the LQR approach have been introduced, resulting in enhanced tracking control performance while addressing issues related to actuator effectiveness loss, parametric uncertainties, and internal disturbances [10]. A robust integral of the signum error was proposed to design an adaptive law using immersion and invariance, successfully achieving tracking control while mitigating unwanted uncertainties and disturbances [11]. However, as an online estimation technique, adaptive control can fail if the quadrotor states are inaccurately measured [12].

Sliding mode control (SMC) is inherently resistant to a wide range of disturbances, as well as parametric and modeling uncertainties [13]. A nonlinear sliding manifold based on tracking error was designed to develop a control law that ensures the quadrotor UAV follows the desired trajectory [14]. This technique also utilized adaptive laws for tuning control gains, demonstrating insensitivity to variations in mechanical configurations. SMC techniques have been developed for attitude and trajectory tracking control objectives, following a two-step procedure that involves defining the manifold and constructing the control law [15–17]. However, these control techniques achieve asymptotic stability and are prone to chattering due to the discontinuous switching function, which can lead to heating issues and damage the actuators, potentially altering the dynamics of the quadrotor model. To minimize chattering, high-order SMC techniques have been proposed, where the discontinuous control inputs are distributed across higher-order terms derived from the higher derivatives of the sliding mode manifold [18, 19]. To achieve finite-time stability, a non-singular terminating SMC technique has been presented in [20]. Although finite-time stability has been achieved, robustness against uncertainties and disturbances has been ensured, and chattering minimized in this research, the issue has not been completely resolved.

The backstepping control (BSC) technique has been developed for quadrotors to achieve desired control performance while minimizing chattering. BSC techniques have been designed to achieve stabilization and trajectory tracking of quadrotor UAVs [21–23]. In these methods, tracking errors are first defined, followed by the indefinite cancelation of cross-coupling terms using Lyapunov criteria. However, these approaches rely on accurate modeling of the quadrotor system. To address unmodeled dynamics, a neural network was employed alongside the BSC technique to ensure Lyapunov stability [24]. Furthermore, the BSC method was modified to tackle both unmodeled dynamics and parametric uncertainties [25]. Although these techniques have achieved remarkable performance in quadrotor control, the cross-cancelation of indefinite terms has led to increased control

activity. Additionally, the control system exhibited a lack of robustness, necessitating empirical tuning for various scenarios.

Control techniques that combine adaptive laws, sliding mode control (SMC), and backstepping control (BSC) methods have been presented in the literature. A control method was proposed in which the switching gain of the SMC technique is integrated into the backstepping design to minimize chattering [26]. Following this, the adaptive backstepping sliding mode control (BSMC) was developed to enable quadrotors to track predefined trajectories. A controller for the decoupled rotational and translational systems was designed, with the BSMC addressing actuator faults in quadrotors and achieving hovering flight mode [27]. However, these control methods typically achieve only asymptotic stability. To attain fast convergence and finite-time stability of the closed-loop system, a terminating sliding manifold for the SMC combined with the BSC method was designed [28]. However, for control gain tuning, a predefined characteristic equation was employed, which is inadequate during real experiments.

In recent trends, model predictive control (MPC) techniques have been extensively applied to quadrotor control [29, 30]. These MPC methods utilize advanced algorithms to predict the future rotational and translational state variables of the quadrotor and optimize the control inputs over a finite-time horizon. Anticipating future states and managing system constraints are significant advantages of these techniques; however, real-time implementation of MPC is challenging due to unwanted external disturbances with unknown bounds. Neural network-based techniques have been designed for approximating the nonlinear dynamics of both rotational and translational systems, achieving trajectory tracking and disturbance attenuation for disturbances with known bounds [31–34]. However, extensive training data and computational power are required for real-time implementation in quadrotor UAVs. In addition to these methods, other popular control approaches for quadrotor control include fuzzy logic control with rule-based design [35, 36], nonlinear dynamic inversion for the cancelation of highly coupled nonlinear terms [37], and geometric control that directly operates on the special Euclidean group (SE(3)) [38], among others, control techniques in [39] and as summarized in a brief survey in Table 2.

Quadrotor UAVs encounter a range of challenges in control design due to disturbances, parametric and modeling uncertainties, actuator faults, and sensor inefficiencies. To address unwanted internal and external disturbances and achieve control objectives such as hovering and tracking, a disturbance observer (DO) can be integrated with the controller to design disturbance observer-based control (DOBC) [40]. A linear dual DO capable of estimating exogenous disturbances with constant peak-to-peak magnitude and frequency was developed [41]. Additionally, a nonlinear DO for a small-scale indoor quadrotor was designed, in which disturbances and uncertainties were lumped together to establish a maximum constant bound [42]. However, these techniques require an accurate model of the quadrotor, and all states and their rates must be available. In conjunction with controllers to achieve tracking control, DOBC based on sliding mode control (SMC) and backstepping control (BSC) techniques has been designed [15, 43]. However, these methods necessitate not only rotational and translational state measurements but also the rates of change of the disturbance observer.

To address the issue of accurate state measurements, state observers serve as a powerful tool for state estimation. In this approach, the quadrotor's rotational and translational outputs are utilized to estimate all state variables. SMC-based output-feedback control (OFC) system was designed using state estimations and disturbance observers (DO) for trajectory tracking in the presence of disturbances, uncertainties, and input saturations [44]. However, the inevitable chattering issue remained unresolved, and empirical tuning was still required for control gains. An adaptive law was designed and combined with a BSC based OFC technique [45]. Nevertheless, BSC requires repeated derivatives of the state variables, which can lead to an "explosion of complexity" due to high control action. Neural networks were employed to approximate partially unknown quadrotor dynamics by combining both rotational and translational systems, with all control inputs defined simultaneously [34]. Additionally, an optimal OFC technique was developed, integrated with a disturbance observer, where all state variables and their rates were replaced with estimated states obtained through a traditional state observer [46]. However, this method assumed a predefined maximum bound on the disturbance, which increased the risk of instability when disturbances exceeded that bound.

Based on the preceding literature review and the brief survey of state estimation methods, a significant research gap exists in addressing the quadrotor control problem. The uncertain environment, disturbances, modeling, and parametric uncertainties, along with the underactuated and strongly coupled nonlinear dynamics of quadrotor UAVs, render the design of trajectory tracking control algorithms considerably more challenging. Therefore, this research aims to provide advancements in addressing these challenges. The major contributions of this paper are as follows:

1. This paper investigates a novel approach to negate the assumptions of full state measurements and complete knowledge of quadrotor dynamics. A state estimation criterion based on the high-gain observer (HGO) is designed and integrated with disturbance observer-based control (DOBC) to formulate an algorithm aimed at achieving nonlinear trajectory tracking objectives.

**Table 2** Brief Survey of State Estimation Methods for Quadrotor UAVs

| Research | Method | Contributions | Limitations |
|---|---|---|---|
| [18, 47] | Sliding Mode Observer | Robust to disturbances and uncertainties | Chattering issue |
| [48–50] | Kalman Filters | Stochastic approach; optimal estimation for linearized quadrotor systems | Assumes noise must be Gaussian; sensitive to modeling uncertainties |
| [51, 52] | Unscented Kalman Filter | Does not rely on linearization; improved performance in estimating nonlinearities | Computational and implementation complexity |
| [51, 53, 54] | Extended Kalman Filter | Sensor fusion; effective for nonlinear systems, fault detection with adaptive threshold | Linearization errors |
| [55, 56] | High-Gain Observer | Fast convergence | Sensitivity to noise; peaking phenomenon due to finite escape time |

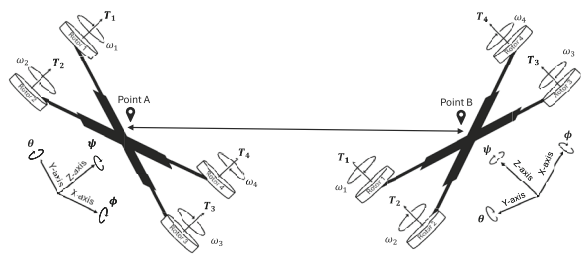


**Fig. 2** Quadrotor movement

2. Dynamic surface control is incorporated within the backstepping control (BSC) framework to address the issue of repeated derivatives in both the controller and the disturbance observer.
3. An adaptive filter is developed based on an auxiliary adaptive control scheme, enabling the filter to adjust to unknown bounded uncertainties and eliminating the need for manual tuning of control gains.
4. A disturbance observer is designed to attenuate nonlinear disturbances, specifically addressing von Kármán turbulence and noises modeled by Gaussian distribution and uniformly random distribution..

The structure of the paper is as follows: Section 2 presents the mathematical modeling and preliminaries necessary for the control design. Section 3 outlines the control design and stability analysis, followed by the control validation and simulations in Section 4. Finally, the paper concludes in Section 5.

## 2 Preliminaries

### 2.1 Problem statement

Fig. 2 illustrates the position and orientation changes of a quadrotor while performing trajectory tracking along a linear path. Even for relatively simple flight tasks, the rotors in the propulsion system must respond rapidly and precisely to generate the required thrust, enabling the quadrotor to change direction and maintain stability. A fundamental aspect of controller design involves minimizing the error between the actual and desired positions. However, achieving precise trajectory tracking becomes challenging in the presence of nonlinear disturbances that affect the quadrotor's dynamics. To address this challenge, this research utilizes a nonlinear error formulation between the desired trajectory and the quadrotor's actual position to develop an effective control strategy. The aim is to achieve trajectory tracking of highly aggressive and dynamic trajectories, ensuring rapid response and stability under complex flight conditions.

Building on this premise, this paper proposes an adaptive filter-based dynamic surface control approach to ensure robust trajectory tracking under multiple nonlinear disturbances in the Cartesian plane. The proposed method begins with the development of an auxiliary control law featuring an adaptive gain to enhance system adaptability. Subsequently, an adaptive filtering mechanism is designed to mitigate the issue of repeated differentiation inherent in the backstepping control technique. A comprehensive control algorithm is then formulated to simultaneously achieve trajectory tracking and disturbance attenuation. To further improve performance, a nonlinear disturbance observer is integrated to estimate and reject external disturbances. Additionally, state estimation is enhanced using a high-gain observer, which is incorporated into the control scheme to formulate a disturbance observer-based control framework. The proposed approach aims to enhance the quadrotor's stability and precision in trajectory tracking, even in the presence of complex environmental perturbations.

### 2.2 Mathematical model

The dynamical model of quadrotor is written as follows [1, 57]:

$$
\begin{aligned}
\ddot{x} &= (\cos\phi\sin\theta\cos\psi + \sin\phi\sin\psi)\,U_p/m \\
\ddot{y} &= (\cos\phi\sin\theta\sin\psi - \sin\phi\cos\psi)\,U_p/m \\
\ddot{z} &= (\cos\phi\cos\theta)\,U_p/m - g \\
\ddot{\phi} &= \dot{\theta}\dot{\psi}\left(I_y - I_z\right)/I_x + \dot{\theta}\Omega_r I_r/I_x + l/I_x U_\phi \\
\ddot{\theta} &= \dot{\phi}\dot{\psi}\left(I_z - I_x\right)/I_y - \dot{\phi}\Omega_r I_r/I_y + l/I_y U_\theta \\
\ddot{\psi} &= \dot{\phi}\dot{\theta}\left(I_x - I_y\right)/I_z + 1/I_z U_\psi
\end{aligned}
\tag{1}
$$

where $m$ and $g$ denote mass of quadrotor and gravitational forces, respectively. $\phi$, $\theta$, and $\psi$ denote attitude angles, i.e., roll, pitch, and yaw, respectively. Furthermore, $l$ is a length of rotor from center of gravity, $I_x$, $I_y$, and $I_z$ denote inertia of quadrotor, and $I_r$ denotes inertia of a rotor. In addition, $\Omega_r = \omega_1 - \omega_2 + \omega_3 - \omega_4$. Next, the control inputs are written as follows:

$$
\begin{aligned}
&U_\phi = b\left(-\omega_2^2 + \omega_4^2\right);\ U_\theta = b\left(-\omega_1^2 + \omega_3^2\right); \\
&U_\psi = d\left(\omega_1^2 - \omega_2^2 + \omega_3^2 - \omega_4^2\right); \\
&U_p = b\left(\omega_1^2 + \omega_2^2 + \omega_3^2 + \omega_4^2\right)
\end{aligned}
\tag{2}
$$

where $b$ and $d$ are denoting lift and resistance coefficients, respectively. Next, assuming $\phi = x_1$, $\dot{\phi} = x_2$, $\theta = x_3$, $\dot{\theta} = x_4$, $\psi = x_5$, $\dot{\psi} = x_6$, $x = x_7$, $\dot{x} = x_8$, $y = x_9$, $\dot{y} = x_{10}$, $z = x_{11}$, and $\dot{z} = x_{12}$. Then, a state-space model can be derived as follows:

$$
\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \dot{x}_4 \\ \dot{x}_5 \\ \dot{x}_6 \\ \dot{x}_7 \\ \dot{x}_8 \\ \dot{x}_9 \\ \dot{x}_{10} \\ \dot{x}_{11} \\ \dot{x}_{12} \end{bmatrix}
=
\begin{bmatrix}
x_2 \\
x_4 x_6\left(\frac{I_y - I_z}{I_x}\right) + x_4\Omega_r\frac{I_r}{I_x} + \frac{l}{I_x}U_\phi \\
x_4 \\
x_2 x_6\left(\frac{I_z - I_x}{I_y}\right) - x_2\Omega_r\frac{I_r}{I_y} + \frac{l}{I_y}U_\theta \\
x_6 \\
x_2 x_4\left(\frac{I_x - I_y}{I_z}\right) + \frac{1}{I_z}U_\psi \\
x_8 \\
\frac{U_x}{m}U_p \\
x_{10} \\
\frac{U_y}{m}U_p \\
x_{12} \\
(\cos x_1\cos x_3)\,\frac{U_p}{m} - g
\end{bmatrix}
\tag{3}
$$

where

$$
\begin{aligned}
U_x &= \cos\phi\sin\theta\cos\psi + \sin\phi\sin\psi \\
U_y &= \cos\phi\sin\theta\sin\psi - \sin\phi\cos\psi
\end{aligned}
\tag{4}
$$

### 2.3 Lemmas and assumptions

**Lemma 1** *1 [58]: According to the continuous property of functions, a continuous function $w(x)$ can be bounded by a maximum value $\bar{w}$ if the initial conditions of the functions are in the compact set.*

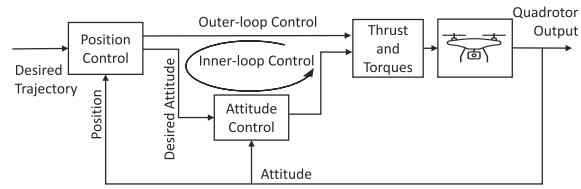


**Fig. 3** Block diagram

**Assumption 1** [12]: Assuming small angle approximation, i.e., $\cos x \approx 1$ and $\sin x \approx x$, (1) can be decoupled into position and attitude submodels consisting of $[\ddot{x}\ \ \ddot{y}\ \ \ddot{z}]^T$ and $[\ddot{\phi}\ \ \ddot{\theta}\ \ \ddot{\psi}]^T$, respectively.

**Assumption 2** The dynamics of attitude control should be much faster than position control [41]. Thus, it is assumed that all the disturbances, faults, modeling, and parametric uncertainties are lumped together and appear in the attitude model.

**Assumption 3** The reference trajectory is locally differentiable.

## 3 Control design and stability analysis

From the mathematical model presented in (1), it is evident that the position dynamics are influenced by the attitude dynamics. Consequently, the model must be decoupled into separate position and attitude components, necessitating the design of an outer loop and an inner loop control scheme. Therefore, controllers for each submodel will be developed in the subsequent sections. As illustrated in Fig. 3, attitude control constitutes the inner loop; thus, a controller for the attitude submodel will be designed first, followed by the development of the position control strategy.

### 3.1 Attitude Control

A dynamical model of a roll submodel subject to unknown bounded disturbances is written as follows:

$$
\ddot{\phi} = \dot{\theta}\dot{\psi}\left(\frac{I_y - I_z}{I_x}\right) + \dot{\theta}\Omega_r\frac{I_r}{I_x} + \frac{l}{I_x}U_\phi + d_\phi(t)
\tag{5}
$$

where $d_\phi(t)$ is a disturbance. Next, a class of state-space model is given by:

$$
\begin{aligned}
\dot{x}_1 &= x_2 \\
\dot{x}_2 &= f_1(x,t) + b_1 U_\phi + d(t)
\end{aligned}
\tag{6}
$$

where

$$f_1(x,t) = x_4 x_6 \left(\frac{I_y - I_z}{I_x}\right) + x_4 \Omega_r \frac{I_r}{I_x};$$
$$b_1 = \frac{l}{I_x}; d(t) = d_\phi(t) \tag{7}$$

Now, the tracking error is defined as follows:

$$z_1 = x_1 - x_{\phi r} \tag{8}$$

where $x_r$ defined for the time interval $[0, \infty)$ represents the tracking reference. Taking the derivative yields:

$$\dot{z}_1 = \dot{x}_1 - \dot{x}_{\phi r} = x_2 - \dot{x}_{\phi r} \tag{9}$$

First-order Levant differentiator to obtain $\dot{x}_{\phi r}$ is designed as follows [59]:

$$\dot{\chi}_1^c = -\kappa_1 |\chi_1^c - x_{\phi r}|^{\frac{1}{2}} \text{sign}\left(\chi_1^c - x_{\phi r}\right) + \chi_{2\phi}^c$$
$$\dot{\chi}_2^c = -\kappa_2 \text{sign}\left(\chi_1^c - x_{\phi r}\right); \tag{10}$$

where $\kappa_1$ and $\kappa_2$ are positive design constants. And $x_{\phi r} \approx \chi_1^c$ is the output and $\chi_2^c$ is its approximated differentiation. Next, an auxiliary control law is chosen as follows:

$$\mu_1 = -p_1 z_1 + \chi_2^c \tag{11}$$

where $p_1 > 0$ is a constant. Next, a filter is then designed as follows:

$$\tau \dot{\xi}_1(t) + \xi_1(t) = \mu_1; \quad \text{where} \quad \xi_1(0) = \mu_1(0) \tag{12}$$

where $\tau$ is a time constant and $\xi_1$ denotes the filter. Next, an error is defined as $e = \xi_1 - \mu_1$. Then, $z_2$ is chosen as follows:

$$z_2 = x_2 - \xi_1 - \chi_2^c \tag{13}$$

Utilizing (13), (9) can be simplified as follows:

$$\dot{z}_1 = -p_1 z_1 + z_2 + e + \chi_2^c \tag{14}$$

Derivative of (13) yields

$$\dot{z}_2 = f_1(x,t) + b_1 U_\phi + d(t) - \dot{\chi}_2^c - \frac{\mu_1}{\tau} + \frac{\xi_1}{\tau} \tag{15}$$

A Lyapunov candidate function can be chosen as follows:

$$V_A = \frac{1}{2} z_1^2 + \frac{1}{2} z_2^2 + \frac{1}{2} e^2 + \frac{1}{2\alpha_\phi} \tilde{p}_1^2 + \frac{1}{2} \tilde{d}_\phi^2 \tag{16}$$

where $\alpha_\phi$ is an adaptive control gain. Additionally, $\tilde{p}_1 = p_1 - \hat{p}_1$ and $\tilde{d}_\phi = \hat{d}_\phi - d_\phi$ where $\hat{p}_1$ is an adaptive gain, and $\hat{d}_\phi$ is the estimation of disturbance obtained according to the DO proposed by [60], written as follows:

$$\dot{\gamma}_\phi = -l_\phi(x) g_{2\phi} \gamma_\phi - l_\phi(x)\big(g_{2\phi} p_\phi(x) + h_\phi(x) + g_{1\phi} U_\phi\big)$$
$$\hat{d}_\phi = \gamma_\phi + p_\phi(x) \tag{17}$$

where $p(x)$ is a nonlinear design function. $l(x)$, $h_\phi(x)$, $g_{1\phi}(x)$ and $g_{2\phi}(x)$ are obtained as follows:

$$l_\phi(x) = \frac{\partial p_\phi(x)}{\partial x}; \quad h_\phi(x) = \left[x_2 \;\; f(x,t)\right]^T;$$
$$g_{1\phi} = \left[0 \;\; b_1\right]^T; \quad g_{2\phi} = \left[0 \;\; 1\right]^T \tag{18}$$

Next, taking the derivative of (41) yields

$$\dot{V}_A = z_1 \dot{z}_1 + z_2 \dot{z}_2 + e\dot{e} + \frac{1}{\alpha}\left(-\dot{\hat{p}}_1\right) \tilde{p}_1 + \tilde{d}_\phi \dot{\tilde{d}}_\phi \tag{19}$$

where $\dot{\tilde{d}}_\phi = \dot{\hat{d}}_\phi - \dot{d}_\phi$ can be obtained as follows:

$$\dot{\tilde{d}}_\phi = -l_\phi(x) g_{2\phi} \tilde{d}_\phi = -\lambda_\phi \tilde{d}_\phi \tag{20}$$

Now, substituting (14), (15), (20) and using $e = \xi_1 - \mu_1$ into (19) yields:

$$\dot{V}_A = -p_1 z_1^2 + z_1 z_2 + z_1 e + z_1 \chi_2^c + z_2\Big(f_1(x) + b_1 U_\phi + d_\phi - \dot{\chi}_2^c - \frac{\mu_1}{\tau} + \frac{\xi_1}{\tau}\Big) - \frac{e^2}{\tau} - F_A(\star) e - \frac{\dot{\hat{p}}_1}{\alpha_\phi} \tilde{p}_1 - \lambda_\phi \tilde{d}_\phi^2 \tag{21}$$

where $F_A(\star) = F_A(x_1, x_2, z_1, z_2, e) = \frac{\partial \mu_1}{\partial x_1} \dot{x}_1 + \frac{\partial \mu_1}{\partial z_1} \dot{z}_1$ is a continuous function. Now, according to Lemma 1, by utilizing the continuous property, the function $F_A(\star)$ can be bounded by a maximum value $\bar{F}_A$, provided that the initial conditions are in the compact set [58]. Subsequently, a controller with an adaptive filter and gain is designed as follows:

$$U_\phi(t) = -\frac{1}{b_1}\Big(z_1 + f_1(x,t) - \dot{\chi}_2^c - \frac{1}{\tau}\left(\hat{\mu}_1 + \xi_1\right) + k_1 z_2 + \hat{d}_\phi(t)\Big);$$
$$\hat{\mu}_1(t) = -\hat{p}_1 z_1 + \chi_2^c; \quad \dot{\hat{p}}_1(t) = -\frac{\alpha_\phi}{\tau} z_1 z_2;$$
$$\dot{\xi}_1(t) = \frac{1}{\tau}\left(\hat{\mu}_1(t) - \xi_1(t)\right) \tag{22}$$

Substituting (22) into (21) and using Young's inequality yields:

$$\dot{V}_{\mathrm{A}} \leq -p_1 z_1^2 - k_1 z_2^2 - \left(\lambda_\phi - \frac{1}{2}\right)\tilde{d}_\phi^2 + \frac{1}{2}\left(z_1^2 + z_2^2\right) + \frac{1}{2}B_1 \tag{23}$$

where $B_1 \geq \max\{\left(e + \dot{x}_{\phi r}\right)^2 - \left(\frac{2}{\tau} + 1\right)e^2 + \bar{F}_{\mathrm{A}}^2\}$. Since, $p_1 > 0$ is an unknown positive constant, by choosing $k_1 > 0$ sufficiently large and designing $l_\phi(x)$ such that $l_\phi(x) g_{2\phi}(x) > \frac{1}{2}$, it can be ensured that $\dot{V}_{\mathrm{A}} < 0$, resulting in an asymptotic stable closed-loop system.

Next, for the design of the OFC algorithm, it is assumed that $d_\phi(t) = 0$ is assumed. Consequently, a high-gain observer (HGO) for state estimation can be designed as follows [61]:

$$\begin{aligned} \dot{\hat{x}}_1 &= \hat{x}_2 + \frac{\beta_{\phi_1}}{\varepsilon}\left(y_\phi - \hat{x}_1\right) \\ \dot{\hat{x}}_2 &= \hat{f}_1(\hat{x}, t) + b_1 U_\phi + \frac{\beta_{\phi_2}}{\varepsilon}\left(y_\phi - \hat{x}_1\right) \end{aligned} \tag{24}$$

where $y_\phi = x_1$ is the measurable output angle, $\hat{f}_1(\hat{x}, u)$ is the nominal model of $f_1(x, t)$, and $\varepsilon > 0$ is a constant. Furthermore, $\beta_{\phi_1}$ and $\beta_{\phi_2}$ are constants such that the polynomial $w^2 + \beta_{\phi_1} w + \beta_{\phi_2}$ is Hurwitz. To perform the stability analysis, the state estimation errors are defined as follows:

$$e_1 = \frac{1}{\varepsilon}\tilde{x}_1; \; e_2 = \tilde{x}_2 \tag{25}$$

Let $e = [e_1, e_2]^T$. Taking derivative and simplifying yields $\varepsilon\dot{e} = Ae + \varepsilon\delta$, where matrices $\delta$ and $A$ are written as follows:

$$A = \begin{bmatrix} -\beta_{\phi_1} & 1 \\ -\beta_{\phi_2} & 0 \end{bmatrix}; \delta = \begin{bmatrix} 0 \\ \tilde{f} \end{bmatrix} \tag{26}$$

where $\tilde{f} = f_1(x, t) - \hat{f}_1(\hat{x}, t)$ and $A$ is Hurwitz due to the characteristic polynomial defined by $\beta_{\phi_1}$ and $\beta_{\phi_2}$. Now, a Lyapunov candidate function is chosen as follows:

$$V_H = e^T P e \tag{27}$$

where $P = [p_{11}, p_{12}; p_{21}, p_{22}]$ with positive constants such that $P = P^T > 0$ and $p_{12} = p_{21}$. In addition $P$ is obtained by solving $PA + A^T P = -I$. Next, taking the derivative of $V_H$ and simplifying, along with using Young's inequality yields:

$$\begin{aligned} \dot{V}_H &= -e^T e + 2\varepsilon e^T P \delta \\ &\leq -\left(1 - \frac{\varepsilon^2}{2}\right)e_1^2 - \left(1 - \frac{\varepsilon^2}{2\beta_{\phi_2}^2}\left(\frac{1}{\beta_{\phi_1}} \times \left(1 + \beta_{\phi_2}\right) + \beta_{\phi_1}\right)\right)e_2^2 + \tilde{f}^2 \end{aligned} \tag{28}$$

Hence, asymptotic stability can be achieved by appropriately choosing $\beta_{\phi_1} > 0$, $\beta_{\phi_2} > 0$ and designing $\varepsilon \in (0, 1]$ to be sufficiently small. The asymptotic stability is also investigated and validated in following corollary.

**Corollary 1** *[61]: Assuming that $\hat{f}_1(\hat{x}, t)$ is locally Lipschitz in its arguments then there exists $\varepsilon^* \in (0, 1]$ such that for $0 < \varepsilon < \varepsilon^*$ the estimation errors $\tilde{x}_1 = x_1 - \hat{x}_1$ and $\tilde{x}_2 = x_2 - \hat{x}_2$ of HGO are bounded as follows:*

$$|\tilde{x}_i| \leq max\left\{\frac{b}{\varepsilon^{i-1}}||\tilde{x}(0)||e^{\frac{-at}{\varepsilon}}, \varepsilon^{\rho+1-i}ck\right\} \tag{29}$$

*where $a$, $b$, $c$ and $k$ are positive constants. $i \in [1, 2]$ and $\rho = 2$ is the order of the system.*

In order to summarize the control developed for achieving the desired tracking objectives, following theorem is proposed.

**Theorem** *For a class of nonlinear systems with nonlinear unknown bounded disturbances described by (6), an output-feedback trajectory tracking control strategy based on dynamic surface control, combined with an adaptive filter and DO, can be designed as follows:*

$$\begin{aligned} &\begin{cases} U_\phi = -\frac{1}{b_1}\left(z_1 + f(\hat{x}) - \frac{1}{\tau}(\mu_1 - \xi_1) \right. \\ \qquad \left. + p_2 z_2 + \hat{d}_\phi(t)\right) \\ \mu_1 = -\hat{p}_1 z_1 + \chi_2^c \end{cases} \\ &\begin{cases} \dot{\gamma}_\phi = -l_\phi(\hat{x}) g_{2\phi}\gamma_\phi - l_\phi(\hat{x})\left(g_{2\phi} p_\phi(\hat{x}) \right. \\ \qquad \left. + h_\phi(\hat{x}) + g_{1\phi} U_\phi\right) \\ \hat{d}_\phi = \gamma_\phi + p_\phi(\hat{x}) \end{cases} \\ &\left\{\dot{\xi}_1 = \frac{1}{\tau}(\mu_1(t) - \xi_1(t)); \; \dot{\hat{p}}_1 = -\frac{1}{\tau}\alpha z_1 z_2\right. \\ &\begin{cases} \dot{\hat{x}}_1 = \hat{x}_2 + \frac{\beta_{\phi_1}}{\varepsilon}(y_\phi - \hat{x}_1) \\ \dot{\hat{x}}_2 = \hat{f}_1(\hat{x}, t) + b_1 U_\phi + \frac{\beta_{\phi_2}}{\varepsilon}(y_\phi - \hat{x}_1) \end{cases} \end{aligned} \tag{30}$$

**Proof** A Lyapunov candidate function can be chosen by combining (16) and (27), written as follows:

$$V = V_A + V_H \tag{31}$$

Taking the derivative and substituting (23) and (28) yields

$$\begin{aligned}\dot{V} \leq &- p_1 z_1^2 - k_1 z_2^2 - \left(1 - \frac{\varepsilon^2}{2}\right) e_1^2 \\ &- \left(1 - \frac{\varepsilon^2}{2\beta_{\phi_2}^2}\left(\frac{1}{\beta_{\phi_1}}\left(1 + \beta_{\phi_2}\right) + \beta_{\phi_1}\right)\right) e_2^2 \\ &- \left(\lambda_\phi - \frac{1}{2}\right)\tilde{d}_\phi^2 + \frac{1}{2}\left(z_1^2 + z_2^2\right) + \frac{1}{2}B_1 + \tilde{f}^2\end{aligned} \tag{32}$$

Hence, by designing appropriate positive gains for control, DO and HGO, along with sufficiently small time constant $\tau$ and $\varepsilon$, asymptotic closed-loop stability are achieved. This concludes the proof.

**Corollary 2** *[62]: According to the separation principle, for state estimation, if the HGO gains are designed to be sufficiently large, i.e., if $\varepsilon$ is sufficiently small, then the OFC technique recovers the trajectories of the SFC technique. Hence, the designed controller can be finalized by replacing state variables with estimated state variables obtained using state estimation criteria.*

For pitch and yaw, similar control development approach can be followed.

### 3.2 Position control

The position state-space model represents an underactuated system characterized by a single control input, $U_p$, and three outputs, $x$, $y$, and $z$, as illustrated in equation (3). In order to develop a position controller, an auxiliary control can be design such that $U = [U_x, U_y, U_z]^T = -gz + \frac{T}{m}Rz$ [1]. Therefore, the position model along x-axis is derived as $\dot{x}_7 = x_8$ and $\dot{x}_8 = U_x + d_x(t)$ where $d_x(t)$ is a disturbances. Next, a tracking error is defined as follows:

$$z_7 = x_7 - r_x \tag{33}$$

where $r_x$ represents the desired position along the x-axis. Now, taking the derivative yields:

$$\dot{z}_7 = \dot{x}_7 - \dot{r}_x = x_8 - \dot{r}_x \tag{34}$$

First-order Levant differentiator to obtain $\dot{x}_{\phi r}$ is designed as follows [59]:

$$\begin{aligned}\dot{\chi}_7^c &= -\kappa_7 |\chi_7^c - r_x|^{\frac{1}{2}} \text{sign}\left(\chi_7^c - r_x\right) + \chi_8^c \\ \dot{\chi}_8^c &= -\kappa_8 \text{sign}\left(\chi_7^c - r_x\right);\end{aligned} \tag{35}$$

where $\kappa_7$ and $\kappa_8$ are positive design constants. And $r_x \approx \chi_7^c$ is the output and $\chi_8^c$ is its approximated differentiation. Next, an auxiliary control law is constructed as follows:

$$\mu_4 = -p_4 z_7 + \chi_8^c \tag{36}$$

where $p_4 > 0$ is a constant. The criteria for the filter are introduced as follows:

$$\tau\dot{\xi}_4(t) + \xi_4(t) = \mu_4; \quad \text{where} \quad \xi_4(0) = \mu_4(0) \tag{37}$$

where $\tau$ is a time constant and $\xi_4(t)$ denotes the filter. Additionally, the error is defined as $e_4 = \xi_4 - \mu_4$. Now, $z_8$ is chosen as follows:

$$z_8 = x_8 - \xi_4 - \chi_8^c \tag{38}$$

By using (38), (34) can be simplified as follows:

$$\dot{z}_7 = -p_4 z_7 + z_8 + e_4 + \chi_8^c \tag{39}$$

Derivative of (38) yields

$$\dot{z}_8 = U_x + d_x(t) - \dot{\chi}_8^c - \frac{\mu_4}{\tau} + \frac{\xi_4}{\tau} \tag{40}$$

For the design of the standard BSC technique, a Lyapunov candidate function can be chosen as follows:

$$V_{\text{P}} = \frac{1}{2}z_7^2 + \frac{1}{2}z_8^2 + \frac{1}{2}e_4^2 + \frac{1}{2\alpha_x}\tilde{p}_4^2 + \frac{1}{2}\tilde{d}_x^2 \tag{41}$$

where $\alpha_x$ is a constant, $\tilde{p}_4 = p_4 - \hat{p}_4$, and $\hat{p}_4$ denotes adaptive gain. Additionally, $\tilde{d}_x = \hat{d}_x - d_x$ where $\hat{d}_x$ is the estimation obtained using the DO designed as follows:

$$\begin{aligned}\dot{\gamma}_x &= -l_x(x)g_{2x}(x)\gamma_x - l_x(x)\big(g_{2x}(x)p_x(x) \\ &\quad + h_x(x) + g_{1x}(x)U_x\big) \\ \hat{d}_x &= \gamma_x + p_x(x)\end{aligned} \tag{42}$$

where $p_x(x)$ is a nonlinear design function. $l_x(x)$, $h_x(x)$, $g_{1x}(x)$, and $g_{2x}(x)$ are obtained as follows:

$$\begin{aligned}&l_x(x) = \frac{\partial p_x(x)}{\partial x}; \quad h_x(x) = \begin{bmatrix}x_8 & 0\end{bmatrix}^T; \\ &g_{1x} = \begin{bmatrix}0 & 1\end{bmatrix}^T; \quad g_{2x} = \begin{bmatrix}0 & 1\end{bmatrix}^T\end{aligned} \tag{43}$$

Next, the error dynamics of DO are obtained as $\dot{\tilde{d}}_x = \dot{\hat{d}}_x - \dot{d}_x$, simplified as follows:

$$\dot{\tilde{d}}_x = -l_x(x)g_{2x}\tilde{d}_x = -\lambda_x\tilde{d}_x \tag{44}$$

where $\lambda_x$ is DO control gain.

By invoking the adaptive gain in the auxiliary controller, it can be expressed as $\hat{\mu}_4 = -\hat{p}_4 z_7 + \dot{r}_x$. Next, taking the derivative of (41) yields

$$\begin{aligned}\dot{V}(t) = &z_7\dot{z}_7 + z_8\dot{z}_+ e_4\dot{e}_4 - \frac{1}{\alpha_x}\dot{\hat{p}}_4\tilde{p}_4 \\ &- l_x(x)g_{2x}\tilde{d}_x\end{aligned} \tag{45}$$

Now, substituting (39), (40) and using $e = \xi - \mu_1$ yields

$$\begin{aligned}\dot{V}_{\mathrm{P}} =& - p_4 z_7^2 + z_7 z_8 + z_7 e_4 + z_7 \chi_8^c + z_8 \big(U_x \\ &+ d_x(t) - \dot{\chi}_8^c - \frac{1}{\tau}(\mu_4 + \xi_4) - \hat{d}_\phi + d_\phi\big) \\ &- \frac{e_4^2}{\tau} - F_{\mathrm{P}}(\star) e_4 - \frac{1}{\alpha_x}\dot{\hat{p}}_4 \tilde{p}_4 - \lambda_x \tilde{d}_x\end{aligned} \tag{46}$$

where $F_{\mathrm{P}}(\star) = F_{\mathrm{P}}(x_7, x_8, z_7, z_8, e_4) = \frac{\partial \mu_4}{\partial x_7}\dot{x}_7 + \frac{\partial \mu_4}{\partial z_7}\dot{z}_7$ is a continuous function. Now, introducing to Lemma 1, the function $F_{\mathrm{P}}(\star)$ can be bounded by a maximum value $\bar{F}_{\mathrm{P}}$ provided that the initial conditions are in the compact set. Next, a control scheme can be formulated as follows:

$$U_x = -z_7 + \dot{\chi}_8^c + \frac{1}{\tau}\left(\hat{\mu}_4 - \xi_4\right) - k_4 z_8 + \hat{d}_x(t) \tag{47}$$

where $\hat{d}_x(t)$ is the estimated disturbance.

Substituting (47) into (46) and using Young's inequality yields

$$\begin{aligned}\dot{V}_{\mathrm{P}} =& - p_4 z_7^2 - k_4 z_8^2 - \left(\frac{1}{\tau} z_7 z_8 + \frac{1}{\alpha_x}\dot{\hat{p}}_7\right)\tilde{p}_4 \\ &+ z_7\left(e_4 + \chi_8^c\right) - \frac{e_4^2}{\tau} + z_8 \tilde{d}_x - F_{\mathrm{P}}(\star) e_4 \\ &- \lambda_x \tilde{d}_x\end{aligned} \tag{48}$$

Designing adaptive law as follows:

$$\dot{\hat{p}}_4 = -\frac{1}{\tau}\alpha_x z_7 z_8 \tag{49}$$

Using Young's inequality and substituting (49) into (48) yields:

$$\begin{aligned}\dot{V}_{\mathrm{P}} \le& - p_4 z_7^2 - k_4 z_8^2 - \left(\frac{1}{\tau} z_7 z_8 + \frac{1}{\alpha_x}\dot{\hat{p}}_4\right)\tilde{p}_4 \\ &+ \frac{1}{2} z_7^2 + \frac{1}{2}\left(e_4 + \chi_8^c\right)^2 - \frac{e_4^2}{\tau} + z_8 \tilde{d}_x \\ &+ \frac{1}{2}\bar{F}_{\mathrm{P}}^2 + \frac{1}{2} e_4^2 \\ \le& - p_4 z_7^2 - k_4 z_8^2 - \left(\lambda_\phi - \frac{1}{2}\right)\tilde{d}_\phi^2 \\ &- \left(\frac{2}{\tau} - 1\right) e^2 + \frac{1}{2} z_7^2 + \frac{1}{2} B_{\mathrm{x}}\end{aligned} \tag{50}$$

where $B_{\mathrm{x}} \ge \max\{(e + \dot{r}_x)^2 + \bar{F}_{\mathrm{P}}^2\}$. Hence, by choosing $p_4 > 0$, $k_4 > 0$, and designing $\tau$ such that $\tau < 2$, it can be ensured that $\dot{V}_{\mathrm{P}} < 0$, resulting in the stable closed-loop system. Next, to achieve the state estimation criteria, a HGO can be designed as follows [61]:

$$\begin{aligned}\dot{\hat{x}}_7 &= \hat{x}_8 + \frac{\beta_7}{\varepsilon}\left(y_x - \hat{x}_7\right) \\ \dot{\hat{x}}_8 &= \hat{f}_4(\hat{x}, U_x, t) + \frac{\beta_8}{\varepsilon^2}\left(y_x - \hat{x}_7\right)\end{aligned} \tag{51}$$

where $y_x = x_7$ is output along the x-axis. In addition, $\beta_7 > 0$ and $\beta_8 > 0$ are HGO gains. Furthermore, $\hat{f}_4(\hat{x}, U_x, t) = \cos\hat{x}_1 \sin\hat{x}_3 \cos\hat{x}_5 + \sin\hat{x}_1 \sin\hat{x}_3$. Using a similar approach yields auxiliary controllers $U_y$ and $U_z$ for $y$-axis and $z$-axis, respectively.

**Remark 1** Since position control requires information of estimated attitude angles. Hence, it should be recalled that the inner loop control dynamics, i.e., attitude control, should be much faster than the outer loop, i.e., position control, to achieve the desired stability and trajectory tracking of quadrotor UAV.

Once the auxiliary controllers are obtained then the position control, $U_p$ can be formulated as follows:

$$U_p = \frac{m\,(U_z + g)}{\cos\phi_{\mathrm{des}}\cos\theta_{\mathrm{des}}} \tag{52}$$

where the desired attitude angles can be obtained by using (4) with $\psi_{\mathrm{des}}$ as a desired known constant yaw angle. Simplifying yields $\theta_{\mathrm{des}} = \theta_{\mathrm{des}} = \tan^{-1}\left(\frac{U_x \cos\psi_{\mathrm{des}} + U_y \sin\psi_{\mathrm{des}}}{U_z + g}\right)$ and $\phi_{\mathrm{des}} = \tan^{-1}\left(\frac{(U_x \sin\psi_{\mathrm{des}} - U_y \cos\psi_{\mathrm{des}})\cos\theta_{\mathrm{des}}}{U_z + g}\right)$.

The complete control flow and algorithm designed in this research paper are summarized in Fig. 4 and algorithm 1, respectively.

## 4 Simulations

To demonstrate and validate the effectiveness of the proposed control approach, simulations were conducted for trajectory tracking along a nonlinear path in the presence of unknown bounded disturbances. The simulations were performed on a real DJI F450 quadrotor model within the MATLAB/Simulink environment. The physical parameters of the quadrotor UAV are listed in Table 3. Additionally, to further illustrate the controller's effectiveness and superiority, a comparison is made with the control method presented in [40].

### 4.1 Simulation setup and environment configuration

#### 4.1.1 Desired trajectories design

Simulations were conducted for three distinct trajectories: a highly aggressive and dynamic trajectory, a helical tra-

**Fig. 4** Control flow

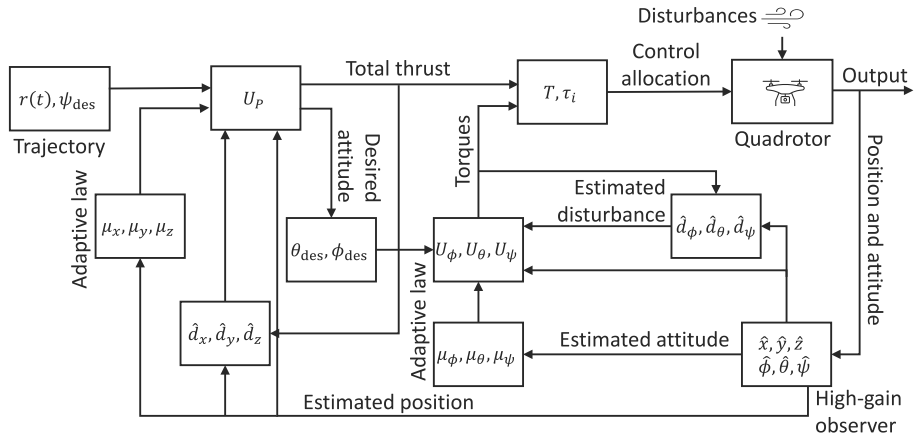


**Table 3** DJI F450 quadrotor parameters [1, 57]

| Parameter | Symbol | Value | Unit |
|---|---|---|---|
| Gravity | $g$ | 9.81 | $m/s^2$ |
| Mass | $m$ | 2 | kg |
| Length | $l$ | 0.225 | m |
| Thrust coefficient | $b$ | $9.86 \times 10^{-6}$ | $Ns^2$ |
| Drag coefficient | $d$ | $1.6 \times 10^{-7}$ | $Nms^2$ |
| Rotor inertia | $J_r$ | $2.8 \times 10^{-6}$ | $kgm^2$ |
| Airframe inertia of roll | $I_x$ | 0.0035 | $kgm^2$ |
| Airframe inertia of pitch | $I_y$ | 0.0035 | $kgm^2$ |
| Airframe inertia of yaw | $I_z$ | 0.0050 | $kgm^2$ |

jectory, and a square trajectory within the Cartesian plane. The helical and square trajectories exhibit uniform behavior along the XYZ axes, whereas the highly aggressive trajectory incorporates aggressive maneuvers. The desired reference trajectory is represented by $r(t) = [x_{\text{des}}, y_{\text{des}}, z_{\text{des}}]^T$. The highly aggressive and dynamic trajectory is divided into two phases, i.e., takeoff-aggressive maneuver phase and landing phase.

The first phase is defined for $0 \leq t < t_f$ written as follows:

$$\begin{aligned} x_{\text{des}}(t) =& \frac{5}{T_s}t - 1.5\sin\left(\frac{5\pi t}{T_s}\right) + \cos\left(\frac{10\pi t}{T_s}\right) \\ y_{\text{des}}(t) =& \frac{5}{T_s}t - 1.5\cos\left(\frac{5\pi t}{T_s}\right) + \sin\left(\frac{10\pi t}{T_s}\right) \\ z_{\text{des}}(t) =& 5\sin^2\left(\frac{\pi t}{T_s}\right) + \sin\left(\frac{10\pi t}{T_s}\right) \end{aligned} \tag{53}$$

The landing phase is defined for $t_f \leq t \leq T_s$ written as follows:

$$\begin{aligned} x_{\text{Landing}}(t) =& \varsigma(k) x_{\text{des}}(t) \\ y_{\text{Landing}}(t) =& \varsigma(k) y_{\text{des}}(t) \\ z_{\text{Landing}}(t) =& \varsigma(k) z_{\text{des}}(t) \end{aligned} \tag{54}$$

where $T_s$ and $T_l$ denote total flight time and landing phase time, respectively. $t_f = T_s - T_l$, $\varsigma(k) = 6(1-k)^5 - 15(1-k)^4 + 10(1-k)^3$ and $k = \frac{t - t_f}{T_s - t_f}$. The remaining two trajectories are given in appendix *I*.

#### 4.1.2 Disturbances

The quadrotor model is subjected to bounded, unknown disturbances and measurement noise. In this study, external aerodynamic disturbances opposing the flight dynamics are modeled using the Von Kármán turbulence model, as detailed in Algorithm 2. This turbulence model provides a more realistic representation of atmospheric disturbances, as it explicitly depends on the quadrotor's altitude and airspeed, both of which influence the intensity and scale of turbulent fluctuations. Furthermore, the model accounts for disturbances along all three translational axes, thereby enhancing the fidelity of numerical simulations by replicating real-world flight conditions more accurately.

Additionally, the noises appearing in the internal attitude dynamics are assumed as uniformly distributed random noise, Gaussian-distributed random noise characterized by a peak-to-peak magnitude range of $[-1, 1]$, a seed of 0, and a sample time of 1 second. Furthermore, band-limited white noise is introduced with a sample time of 0.5 seconds, a seed of [23341], and noise power set to 0.1. These disturbances are generated using Simulink built-in blocks.

### 4.2 Results

#### 4.2.1 Trajectory tracking

After conducting simulations for the DJI F450 quadrotor UAV, remarkable trajectory tracking control performance has been achieved for all of the four trajectories. The control parameters required to achieve the desired control performance are presented in Table 4, where $i \in (\phi, \theta, \psi, x, y, z)$ and $j \in (1, 3, 5, \ldots, 11)$. In all cases, the quadrotor is ini-

**Algorithm 1** Control algorithm

**Require:** Quadrotor dynamics, parameters, and reference trajectory: $r(t)$
**Ensure:** Error $\rightarrow 0 : x_{\text{des}} \leftarrow x(t),\ y_{\text{des}}(t) \leftarrow y(t),\ z_{\text{des}}(t) \leftarrow z(t)$ ▷ *Tracking*
$0 < \varepsilon < 1, 0 < \tau < 2, u(t) = 0, \hat{d}(t) = 0, \psi_{\text{des}}, \phi(0), \theta(0), \psi(0), x(0), y(0), z(0)$ ▷
**while** Tracking error $\neq 0$ **do**
  **for** $i \in (1, 3, 5, \ldots, 11)$ **do**
    $\dot{\hat{x}}_i = \hat{x}_{i+1} + \frac{\beta_i}{\varepsilon}\left(x_i - \hat{x}_i\right);$
    $\dot{\hat{x}}_{i+1} = \hat{f}_{\frac{1}{2}(i-1)+1} + \frac{\beta_{i+1}}{\varepsilon^2}\left(x_i - \hat{x}_i\right)$
    $\dot{\chi}_i^c = -\kappa_i|\chi_i^c - r(t)|^{\frac{1}{2}}\text{sign}\left(\chi_i^c - r(t)\right)$
    $+\chi_{i+1}^c$
    $\dot{\chi}_{i+1}^c = -\kappa_{i+1}\text{sign}\left(\chi_i^c - r(t)\right);$
    $i = i + 2$
  **end for**
  **if** $\tilde{\phi}(t) \neq 0, \tilde{\theta}(t) \neq 0$ and $\tilde{\psi}(t) \neq 0$ **then**
    **for** $i \in (1, 2, 3)$ and $j \in (\phi, \theta, \psi)$ **do**
      $\hat{\mu}_i = -\hat{p}_i z_{2i-1} + \dot{x}_{jr};$
      $\dot{\xi}_i(t) = \frac{1}{\tau}\left(\hat{\mu}_i - \xi_i\right),$
      $\dot{\hat{p}}_i = -\frac{1}{\tau}\alpha_j z_{2i-1} z_{2i}$
      **while** $\tilde{d}(t) \neq 0$ **do**
        $\dot{\gamma}_j = -l_j g_{(2i)j}\gamma_j - l_j(g_{(2i)j}p_j + h_j$
        $+g_{ij}U_j);\ \hat{d}_j(t) = \gamma_j + p_i(x)$
      **end while**
      $U_j = \frac{-1}{b_i}\left(z_{2i-1} + f_i(\hat{x}_i) - \frac{(\hat{\mu}_i - \xi_i)}{\tau}\right.$
      $\left.+k_i z_{i+1} + \hat{d}_j\right);$
      $i = i + 1$
    **end for**
  **end if**
  **if** $\tilde{x}(t) \neq 0, \tilde{y}(t) \neq 0$ and $\tilde{z}(t) \neq 0$ **then**
    **for** $i \in (4, 5, 6)$ and $j \in (x, y, z)$ **do**
      $\hat{\mu}_i = -\hat{p}_i z_{2i-1} + \dot{r}_j;$
      $\dot{\xi}_i(t) = \frac{1}{\tau}\left(\hat{\mu}_i - \xi_i\right);$
      $\dot{\hat{p}}_i = -\frac{1}{\tau}\alpha_j z_{2i-1} z_{2i}$
      **while** $\tilde{d}(t) \neq 0$ **do**
        $\dot{\gamma}_j = -l_j g_{(2i)j}\gamma_j - l_j(g_{(2i)j}p_j + h_j$
        $+g_{ij}U_j);\ \hat{d}_j(t) = \gamma_j + p_i(x)$
      **end while**
      $U_j = -z_{2i-1} + \ddot{r}_j + \frac{1}{\tau}\left(\hat{\mu}_i - \xi_i - k_i z_{2i}\right)$
      $i = i + 1$
    **end for**
    $\theta_{\text{des}};\ \phi_{\text{des}};\ U_p = \frac{m(U_z + g)}{\cos\phi_{\text{des}}\cos\theta_{\text{des}}}$
  **end if**
  $U_\phi, U_\theta, U_\psi, U_p \rightarrow$ Control allocation
**end while**

**Algorithm 2** Von KÃ¡rmÃ¡n Turbulence

**Given:** $h, V, t, T_{sim}$
**Output:** $\mathbf{W}_{inertial} = [W_x, W_y, W_z]^T$
**Initialize:** $\Delta t = 0.01, t_{vec} = \{0, \Delta t, 2\Delta t, \ldots, T_{sim}\}, N = |t_{vec}|$
$L_x = L_y = L_z = 0.177h^{4/3}$
$\sigma_w = 0.1V$
$\xi_x, \xi_y, \xi_z \sim \mathcal{N}(0, 1), \quad \xi \in \mathbb{R}^{3\times N}$
$W_x(1) = 0, \quad W_y(1) = 0, \quad W_z(1) = 0$
**for** $i = 2$ to $N$ **do**
  $W_x(i) = W_x(i-1) - \frac{W_x(i-1)\Delta t}{L_x} +$
  $\sigma_w\sqrt{\frac{2\Delta t}{L_x}}(1 + 0.2W_x(i-1))\xi_x(i)$
  $W_y(i) = W_y(i-1) - \frac{W_y(i-1)\Delta t}{L_y} +$
  $\sigma_w\sqrt{\frac{2\Delta t}{L_y}}(1 + 0.2W_y(i-1))\xi_y(i)$
  $W_z(i) = W_z(i-1) - \frac{W_z(i-1)\Delta t}{L_z} +$
  $\sigma_w\sqrt{\frac{2\Delta t}{L_z}}(1 + 0.2W_z(i-1))\xi_z(i)$
**end for**
Find Nearest Index: $k = \arg\min_i |t_{vec}(i) - t|$
Output: $\mathbf{W}_{inertial} = \left[W_x(k)\ W_y(k)\ W_z(k)\right]^T$

**Table 4** Control parameters

| Gain | $\phi(t)$ | $\theta(t)$ | $\psi(t)$ | $x(t)$ | $y(t)$ | $z(t)$ |
|---|---|---|---|---|---|---|
| Highly aggressive trajectory | | | | | | |
| $p_j$ | 40 | 15 | 10 | 1 | 1 | 1 |
| $k_j$ | 15 | 15 | 10 | 2 | 2 | 2 |
| $\beta_{i_1}$ | 1 | 1 | 1 | 1 | 1 | 1 |
| $\beta_{i_2}$ | 2 | 2 | 2 | 2 | 2 | 2 |
| Helical trajectory | | | | | | |
| $p_j$ | 40 | 15 | 10 | 50 | 10 | 35 |
| $k_j$ | 15 | 15 | 10 | 2 | 2 | 2 |
| $\beta_{i_1}$ | 1 | 1 | 1 | 1 | 1 | 1 |
| $\beta_{i_2}$ | 2 | 2 | 2 | 2 | 2 | 2 |
| Square trajectory | | | | | | |
| $p_j$ | 40 | 15 | 10 | 15 | 15 | 35 |
| $k_j$ | 15 | 15 | 10 | 2 | 2 | 2 |
| $\beta_{i_1}$ | 1 | 1 | 1 | 1 | 1 | 1 |
| $\beta_{i_2}$ | 2 | 2 | 2 | 2 | 2 | 2 |

tially positioned at the origin, with attitude angles set to zero, specifically $\phi(0) = 0$, $\theta(0) = 0$, and $\psi(0) = 0$.

The performance of the quadrotor during the trajectory tracking in the Cartesian plane is illustrated in Figs. 5, 6, and 7. It has been observed that the controller developed in this research work and the comparative controller designed in [40] exhibit similar control performance for smooth, consistent, and non-aggressive trajectories. However, when exposed to the highly aggressive and dynamic trajectory, and Lissajous trajectory, the comparative controller struggles and yields significant tracking error. In contrast, the controller designed in this research work achieves tracking of aggressive maneuvers effectively, and hence reducing the root mean square errors (RMSE) significantly as compared to the controller in [40], as shown in the RMSE barchart illustrated in Fig. 8.

Figure 9 presents the simulation results for the position, velocity, acceleration, and jerk of the quadrotor during a highly aggressive and dynamic trajectory tracking flight mode. The position plot highlights three distinct phases: takeoff, aggressive maneuvers, and landing. During the takeoff phase, the quadrotor ascends smoothly, demonstrating gradual and controlled motion that ensures stability and minimizes the application of abrupt forces in the early stages

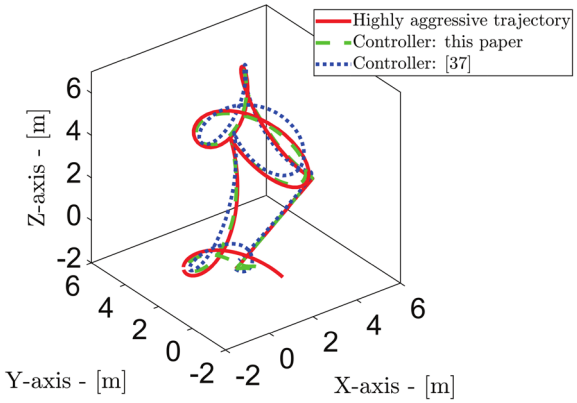


**Fig. 5** Trajectory tracking—Highly aggressive and dynamic trajectory

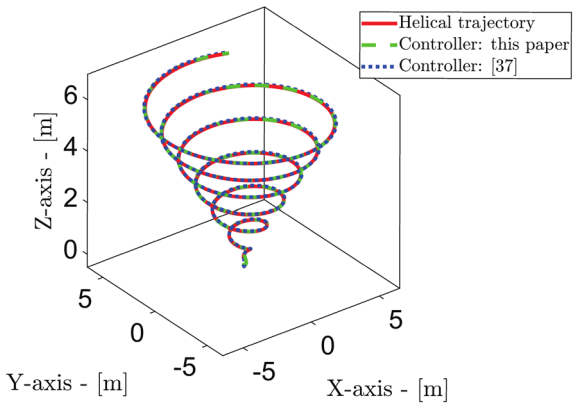


**Fig. 6** Trajectory tracking—Helical trajectory

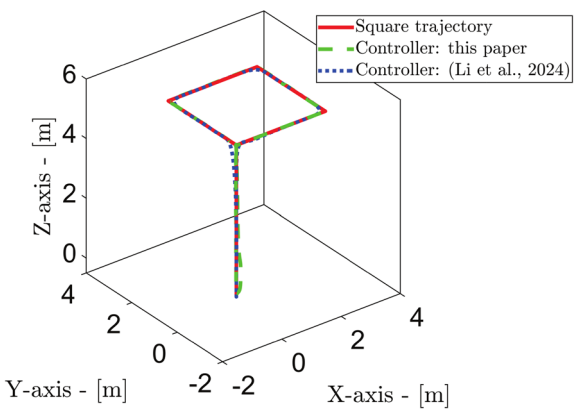


**Fig. 7** Trajectory tracking—Square trajectory

of flight. In the aggressive maneuvering phase, the quadrotor exhibits dynamic motion along the X, Y, and Z axes, reflecting precise control and high responsiveness. During the landing phase, the quadrotor transitions smoothly into a stable descent, underscoring the robustness of the control system in managing complex flight dynamics and disturbances.

Additionally, the velocity, acceleration, and jerk plots provide further insight into the quadrotor's performance. To

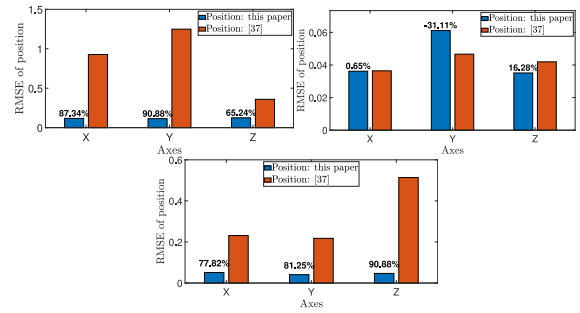


**Fig. 8** RMSE barcharts of trajectory tracking

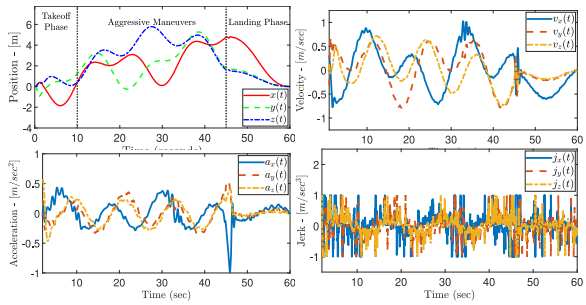


**Fig. 9** Position, velocity, acceleration, and jerk while highly aggressive and dynamic trajectory

execute aggressive maneuvers, the velocity and acceleration adapt dynamically to the trajectory demands. While some chatter is observed in the jerk profile due to external disturbances and noise, it remains within acceptable limits, ensuring the stability of the quadrotor's dynamics throughout the mission. This analysis emphasizes the effectiveness of the control system in maintaining stable and reliable performance under disturbances and noises conditions.

Figure 10 shows the quadrotor's position and attitude tracking with respect to time for highly aggressive trajectory. It is evident that the controller achieves desired tracking control effectively. It is important to highlight that the yaw angle serves as a known reference input provided to the controller, together with the intended trajectory. Hence, the plot of yaw angle is constant unlike other quadrotor attitude and position state variables. Similar tracking performance with respect to time is achieved for other trajectories. Thus, it is omitted here.

During the simulations, it was observed that the controller used for comparison required an empirical method for control gain design, which posed significant challenges in achieving effective tracking of the highly aggressive and Lissajous trajectories due to its inherent sharp maneuvers. In contrast, the proposed controller's reliance on adaptive control laws facilitated a more straightforward design of the control gains. Consequently, the controller presented in this paper is easier to implement and demonstrates superior performance compared to the controller used for comparison.

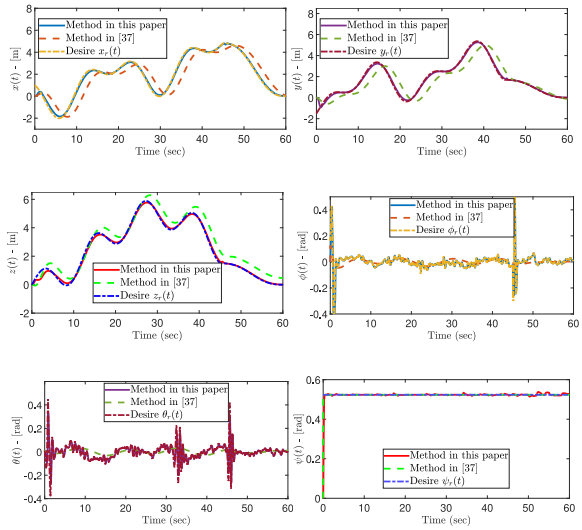


**Fig. 10** Quadrotor outputs during highly aggressive and dynamic trajectory tracking: $[x, y, z, \phi, \theta, \psi]^T$

**Remark 2** Desired roll and pitch angles are generated by using position control whereas yaw angle is initiated as $\psi_{\text{des}} = 30^0 = \pi/6$ radians.

### 4.2.2 Disturbance attenuation

In DOBC algorithms, it is essential for the DO to exhibit faster convergence than the controller itself. This principle is implemented in the current study. To ensure that the DO converges more rapidly than the controller, the following nonlinear disturbance observer function is designed:

$$\begin{bmatrix} p_\phi(t) \\ p_\theta(t) \\ p_\psi(t) \end{bmatrix} = \begin{bmatrix} \lambda_\phi & 0 & \\ 0 & \lambda_\theta & 0 \\ 0 & 0 & \lambda_\psi \end{bmatrix} \begin{bmatrix} x_2 \\ x_4 \\ x_6 \end{bmatrix}; \quad \begin{bmatrix} p_x(t) \\ p_y(t) \\ p_z(t) \end{bmatrix} = \begin{bmatrix} \lambda_x & 0 & \\ 0 & \lambda_y & 0 \\ 0 & 0 & \lambda_z \end{bmatrix} \begin{bmatrix} x_8 \\ x_{10} \\ x_{12} \end{bmatrix} \tag{55}$$

where $\lambda_\phi = \lambda_\theta = \lambda_\psi = 200$ and $\lambda_x = \lambda_y = \lambda_z = 5$ are design gain of DOs.

The results of disturbance attenuation and estimation errors for both the attitude and position submodels are presented in Figs. 11 and 12, respectively. Gaussian-distributed disturbances, uniformly distributed disturbances, and band-limited white noise, which manifest as internal disturbances within the quadrotor's attitude model, are effectively estimated using the roll, pitch, and yaw DOs. Similarly, the quadrotor is subjected to von Kármán turbulence to emulate realistic position disturbances. The simulation results demonstrate that the proposed disturbance observer effectively estimates external perturbations. Although a pronounced peak is observed during the transition from aggressive maneuvers to landing, the observer consistently exhibits robust performance, ensuring accurate disturbance estimation and effective rejection throughout the operational envelope.

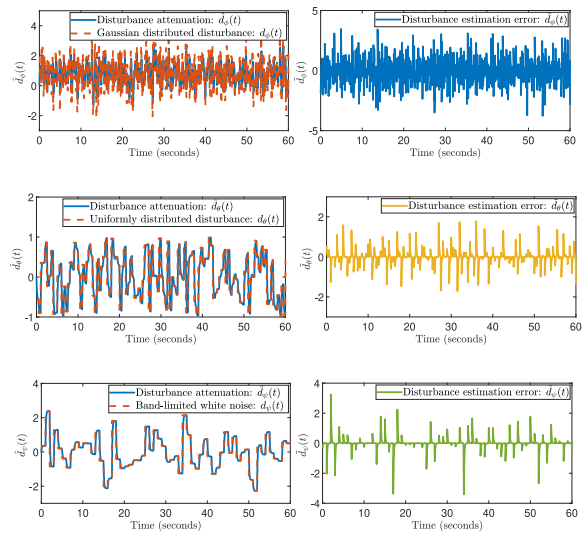


**Fig. 11** Attitude disturbance attenuation and estimation error

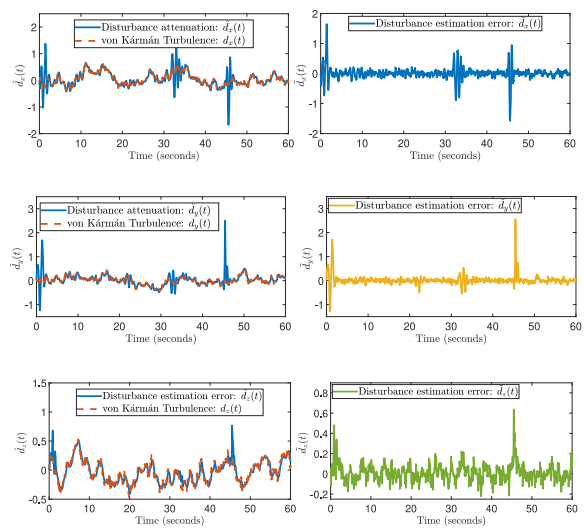


**Fig. 12** Position disturbance attenuation and estimation error

**Remark 3** The proposed controller in this study is based on state estimation; thus, all state variables in the disturbance observers are derived from the estimated state variables obtained using the HGO.

### 4.2.3 State estimation

Figure 13 illustrates the estimated position and attitude angles during the aggressive maneuvers trajectory tracking control.

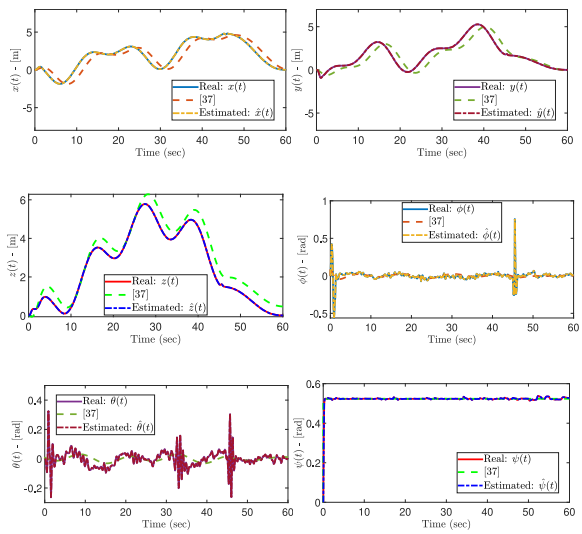


**Fig. 13** Estimated position and attitude in aggressive maneuvers trajectory tracking: $[\hat{x}, \hat{y}, \hat{z}, \hat{\phi}, \hat{\theta}, \hat{\psi}]^T$

For state estimation, the HGO technique has been integrated into the controller, utilizing the measurable quadrotor attitude and position outputs to derive the estimates. Once the estimates of the position and attitude angles are obtained, these estimated state variables and its rates are employed within the controller. Consequently, the implementation of the controller proposed in this paper requires only the desired trajectory, desired yaw angles, and measurable quadrotor position and attitude, thereby eliminating the necessity for measuring velocities using onboard sensors. It is important to note that the convergence of state estimation is influenced by the parameter $\varepsilon$. A smaller value for this parameter facilitates quicker convergence; in the presented simulations, it has been set to $\varepsilon = 10^{-5}$.

#### 4.2.4 Control inputs and filters

Figure 14 illustrates the position and attitude control inputs required to achieve the desired trajectory tracking for the quadrotor UAV. Additionally, the torques and forces exerted by each rotor are presented in Fig. 15. This figure also displays the filters applied to the position and attitude control inputs, addressing the inherent problem of backstepping control, i.e., "explosion of complexity," due to high control inputs.

The variable position control input ensures continuous attenuation of disturbances encountered by the quadrotor during flight missions. When the landing phase begins, a sharp peak occurs due to the abrupt change in trajectory, after which the position control operates smoothly to ensure a precise and seamless landing. Similarly, disturbance compensation is evident in the roll, pitch, and yaw control inputs, illustrating how the thrust and torque are effectively adapted to

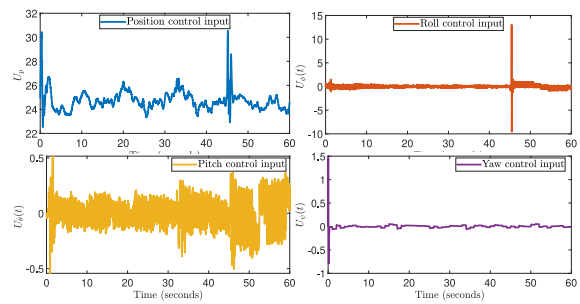


**Fig. 14** Position and attitude control inputs $[U_p, U_x, U_y, U_z]^T$

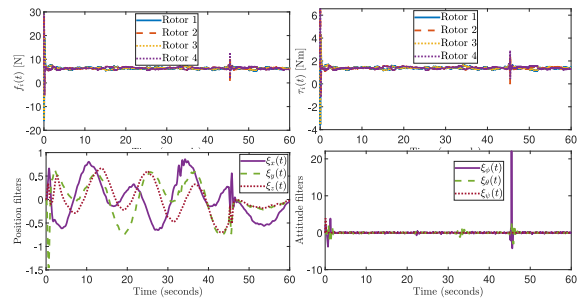


**Fig. 15** Force and torque of each rotor

counteract external disturbances and maintain robust trajectory tracking performance.

#### 4.2.5 Thrust-to-weight ratio, motor dynamics, energy, and power consumption

The maximum forces and torques for each rotor are given by $[f_1, f_2, f_3, f_4] = [19.0982, 28.5504, 19.1322, 29.1366]$ N. Therefore using 1N $\approx$ 101.9716g, the rotor thrusts can be approximated as $[T_1, T_2, T_3, T_4] \approx [1947.3, 2910.2, 1950.1, 2970.3]$ g. Next, the total thrust is obtained as 102.9174N. Therefore, the thrust-to-weight ratio is obtained as $T/W \approx 4.2$. Hence, it indicates that due to excessive thrust, the quadrotor is performing high-performance aggressive maneuvers in the presence of disturbances.

Next, aiming at handling approximately 103N thrust, a motor capable of handling at least 26 N of thrust is required. In this study, the DJI F450 quadrotor is simulated, which is equipped with DJI 2212/920KV motors. However, alternative motor options include EMAX MT2213 935KV Brushless Motor, T-Motor Air 2216 880KV, and SunnySky X2212-980KV II Brushless Motor, with maximum thrust capacities of 1200 g, 1400 g, and 1200 g, respectively. Thus, the thrust values obtained using the controller developed in this paper validate the motor dynamics.

**Remark 4** The position and attitude control inputs increase with a decrease in $\varepsilon$, consequently leading to an increase in the force, torque, and thrust of each rotor. Thus, different

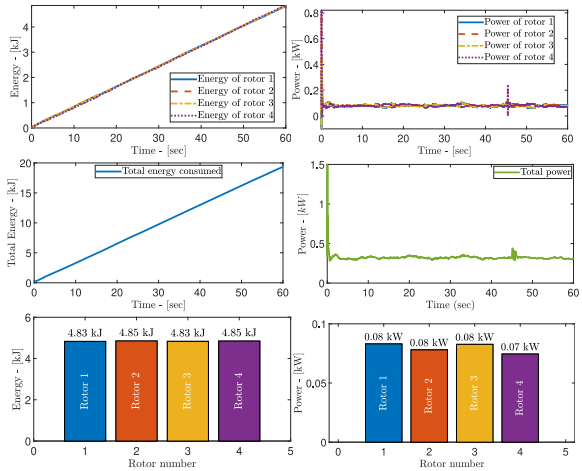


**Fig. 16** Power and energy consumption

motor options may be required depending on the required thrust.

Figure 16 presents the simulation results for the energy and power consumption of each rotor, along with the total energy and power of the quadrotor, during highly aggressive trajectory tracking and maneuvers. The results indicate that all rotors operate with consistent performance, delivering the necessary power to accomplish the flight mission effectively. It is noteworthy that the total power demand at the onset of the simulation is significantly elevated. This is attributed to the initial conditions, where the quadrotor starts from rest at the origin and must rapidly ascend to initiate trajectory tracking.

The total power and energy consumption of the quadrotor throughout the maneuver were observed to be $0.31\,\text{kW} = 310\,\text{W}$ and $19.36\,\text{kJ} = 19360\,\text{J}$, respectively. Considering a 3 S LiPo battery with specifications of 11.1 V and 2200 mAh, the total available energy is calculated as 87120 J. Consequently, the total energy consumption reported in the simulation for a one-minute flight mission remains well within the capacity of the specified battery, thereby affirming the feasibility of the proposed controller.

### 4.3 Discussion on comparison, real experiments, and limitations

In this paper, we present a controller designed to address significant real-time practical challenges associated with unknown disturbances, the reliance on onboard sensors for accurate measurements of quadrotor position and attitude, rotational and translational velocities, and the inherent problem of "explosion of complexity" in backstepping control design. Additionally, adaptive laws are developed to eliminate the need for empirical tuning of control gains. The proposed controller is compared with the one presented by [40], and simulations are conducted on a real DJI F450 quadrotor for three distinct trajectories: the highly aggressive trajectory, the helical trajectory, and the square trajectory.

Both controllers demonstrate impressive performance in tracking the smooth and non-aggressive trajectories. However, the comparative controller exhibits a steady-state error when attempting to track the highly aggressive trajectory. The lack of adaptive laws in the comparative controller complicates the tuning of control gain parameters, whereas the controller proposed in this study achieves precise tracking of the highly aggressive trajectory. The incorporation of adaptive laws simplifies the tuning of control parameters significantly. Furthermore, the nonlinear DO introduced in this research effectively attenuates disturbances with variable frequency and magnitude, as well as uniformly distributed random disturbances and band-limited white noise. In contrast, the comparative controller is limited to attenuating exogenous disturbances with constant frequency and magnitude.

The implementation of the controller developed in this paper necessitates only the desired trajectory and measurable outputs; onboard sensors are solely utilized to measure the position and attitude of the quadrotor, thus eliminating the need for velocity measurements due to the state estimation criteria. The tuning process for the proposed controller is straightforward, relying on adaptive laws, specifically the tuning of $p_i$ in adaptive laws, DSC gain $k_i$, and HGO gains $\frac{\beta_i}{\varepsilon}$. In contrast, the comparative controller lacks these adaptive law criteria and therefore requires extensive control design through empirical tuning, making it a more challenging task when applied across different scenarios. Moreover, it necessitates accurate information regarding position and attitude, along with their rates, for implementation. While quick convergence can be ensured by setting $\varepsilon$ to a sufficiently small value, this approach may lead to increased control actions that could violate the motor dynamics criteria due to excessive thrust.

Despite achieving remarkable trajectory tracking performance for the trajectories, the proposed control technique does not incorporate techniques for handling input delays. In contrast, the comparative controller accounts for input delays; however, its backstepping-based control strategy can result in an "explosion of complexity" due to the recursive differentiation of system states and virtual control signals. The proposed approach in this study addresses this challenge by incorporating filtering techniques.

The controller proposed in this study is based on the HGO, which offers notable advantages in terms of rapid convergence and robustness to high-frequency measurement noise, both of which are essential for high-performance quadrotor control. The application of HGO also facilitates the relaxation of assumptions regarding full state measurements and exact knowledge of the system dynamics. However, effective

operation of the HGO requires accurate measurements of the quadrotor's output position and attitude prior to state and rate estimation. Consequently, any ineffective onboard sensors may result in inaccurate estimations, potentially leading to quadrotor instability. Therefore, the design of a fault-tolerant control system combined with the proposed controller is an interesting direction for future research. Additionally, implementing the controller in an experimental setup, developing neural networks for system dynamics estimation, and establishing criteria for the optimal choice of $\varepsilon$ are identified as significant limitations of the controller proposed in this research.

## 5 Conclusion

In this research paper, a control algorithm based on an adaptive filter, adaptive gain, and dynamic surface control (DSC) integrated with a nonlinear disturbance observer (DO) has been presented to achieve trajectory tracking and attenuation of multiple nonlinear bounded disturbances. To refine the control technique, a state estimation criterion based on a high-gain observer (HGO) has been incorporated into the controller. The implementation of this control algorithm requires only measurable output information regarding the position and attitude of the quadrotor UAV, as well as the desired trajectory, thereby eliminating the need for velocity measurements using onboard sensors. To demonstrate the control effectiveness, stability analysis and simulation studies have been conducted, revealing successful achievement of the quadrotor tracking control objectives for aggressive and non-aggressive trajectories. Based on the control development, stability analysis, and simulation results, it is concluded that the controller is both effective and stable in achieving the desired trajectory tracking control objectives in uncertain environments characterized by multiple disturbances.

## Appendix *I*

### Helical trajectory

$$x_{\text{des}}(t) = \frac{1}{10} + \frac{1}{20} t \sin\left(\frac{t}{\pi}\right)$$
$$y_{\text{des}}(t) = \frac{1}{10} + \frac{1}{20} t \cos\left(\frac{t}{\pi}\right)$$
$$z_{\text{des}}(t) = \frac{1}{5} + \frac{1}{20} t$$

### Square trajectory

The square trajectory is divided into three phases, i.e., takeoff, path following, and landing phase. During takeoff, the trajectory is defined as $x_{\text{des}}(t) = 0$, $y_{\text{des}}(t) = 0$, $z_{\text{des}}(t) = \frac{h}{T_{\text{to}}} t$. Define, $T_{\text{to}}$ is takeoff-time, $T_{\text{s}}$ is time of total segment, $T_{\text{p}}$ denotes time of each side, $t_{\text{p}} = t - T_{\text{to}}$ and $T_{\text{s}} = \frac{T_{\text{p}}}{4}$. And the landing phase is defined as $x_{\text{des}}(t) = 0$, $y_{\text{des}}(t) = 0$ and $z_{\text{des}}(t) = h - \frac{h}{T_{\text{l}}} t_{\text{l}}$ where $T_{\text{l}}$ is landing time and $T_{\text{total}} = T_{\text{to}} + T_{\text{p}} + T_{\text{l}}$is the total flight time. Furthermore, $L$ is the side length of the square and $h$ is maximum height during the trajectory. Therefore, path following trajectory is defined as follows:

$$x_{\text{des}}(t) = \begin{cases} \frac{L}{T_{\text{s}}} t_{\text{p}}, & 0 \le t_{\text{p}} < T_{\text{s}}, \\ L, & T_{\text{s}} \le t_{\text{p}} < 2T_{\text{s}}, \\ L - \frac{L}{T_{\text{s}}}(t_{\text{p}} - 2T_{\text{s}}), & 2T_{\text{s}} \le t_{\text{p}} < 3T_{\text{s}}, \\ 0, & 3T_{\text{s}} \le t_{\text{p}} \le T_{\text{p}}, \end{cases}$$
$$y_{\text{des}}(t) = \begin{cases} 0, & 0 \le t_{\text{p}} < T_{\text{s}}, \\ \frac{L}{T_{\text{s}}}(t_{\text{p}} - T_{\text{s}}), & T_{\text{s}} \le t_{\text{p}} < 2T_{\text{s}}, \\ L, & 2T_{\text{s}} \le t_{\text{p}} < 3T_{\text{s}}, \\ L - \frac{L}{T_{\text{s}}}(t_{\text{p}} - 3T_{\text{s}}), & 3T_{\text{s}} \le t_{\text{p}} \le T_{\text{p}}, \end{cases}$$
$$z_{\text{des}}(t) = h$$

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**Data availability statement** This study did not generate new data. All data used is presented in the paper.

**Code availability statement** No custom code was developed for this study. Analyses were conducted using standard software tools.

## Declarations

**Declarations of interest** None

**Conflict of interest** We have no Conflict of interest to disclose.

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